

BloodLink

Blood Bank & Emergency Blood Coordination System

DBMS Project Proposal Presentation

Database Management Systems (DBMS) | HTML • JavaScript • PHP • MySQL

Team Members: 3 Members

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1. Introduction

BloodLink is a centralized, database-driven web application for managing the complete blood coordination lifecycle.

Core lifecycle: Donor → Donation → Blood Bag → Inventory → Request → Matching → Fulfillment → Audit.

Designed as a role-based relational DBMS with granular Role-Based Access Control (RBAC).

Emphasizes normalized database architecture (3NF), transactional integrity, structured business logic, and automated auditing.

2. Problem Statement & Motivation

Inventory Fragmentation

Hospitals lack real-time visibility into available blood stocks across nearby facilities.

Double Allocation

Concurrent requests can reserve the same limited blood units without strict transactional locking.

Expiry & Wastage

Aggregate stock counts do not track individual collection and expiry dates, increasing spoilage.

Manual Donor Sourcing

Emergency staff may rely on manual calls without dynamic eligibility verification.

Why BloodLink?

Centralized relational data + FEFO + rule-based matching + ACID transactions provide a focused DBMS solution.

3. Project Objectives

- Centralized Lifecycle Tracking: Model the complete blood coordination chain from donor to audit.
- FEFO Inventory Management: Prioritize earliest-expiring blood bags and generate near-expiry alerts.
- Transactional Consistency: Use BEGIN, COMMIT, and ROLLBACK to prevent double allocation.
- Rule-Based Emergency Donor Matching: Identify eligible and compatible donors when inventory is insufficient.
- Comprehensive Audit & Reporting: Maintain historical movement logs and administrative audit trails.

4. Benchmark Study / Existing-System Gap

Fragmented / Spreadsheet-Based

Static or distributed records make regional inventory visibility and unit-level traceability difficult.

Manual Emergency Sourcing

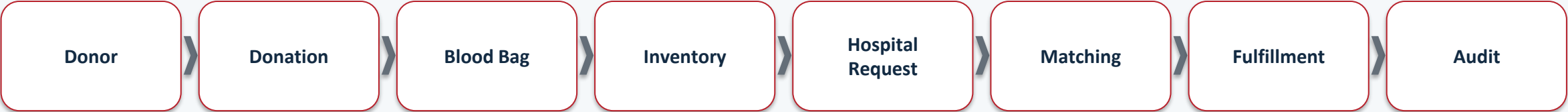
Phone-based donor sourcing lacks automated compatibility and eligibility evaluation.

BloodLink Approach

A centralized 3NF relational model connects donor, donation, bag, inventory, request, matching, fulfillment and audit data.

Benchmark focus in the proposal: replacing fragmented spreadsheets with a centralized, reliable relational database model.

5. Project Overview & System Roles



Donor

Profile, availability, donation history, eligibility date and emergency alerts.

Hospital Staff

Verified access, request creation, real-time request status and regional inventory visibility.

Administrator

Hospital verification, inventory monitoring, reallocation, audit logging and reporting.

6. Donor Functionality

- **Account & Profile Management:** Register an account and maintain contact details, blood group, location, and emergency availability.
- **Donation History:** View previous donation records and check the next eligible donation date based on safety intervals.
- **Emergency Alerts:** Receive in-system alerts for compatible local blood requests when inventory is insufficient.
- **Participation Management:** Accept or decline emergency donation requests directly through the system.
- **Recognition & Rewards:** Track successful donations and recognition tiers — **Bronze, Silver, Gold, and Platinum.**

7. Hospital & Administrator Functionality

Hospital Staff

- Verified account after administrative approval
- Create requests: group, component, quantity, urgency and timeline
- Track PENDING, MATCHING, PARTIALLY_FULFILLED, FULFILLED, CANCELLED
- View regional stock without exposing donor PII

Administrator

- Verify/activate/reject hospital accounts
- Monitor inventory and near-expiry alerts
- Authorize inter-facility reallocations
- Access immutable audit trails and reports

8. Scope of the Project

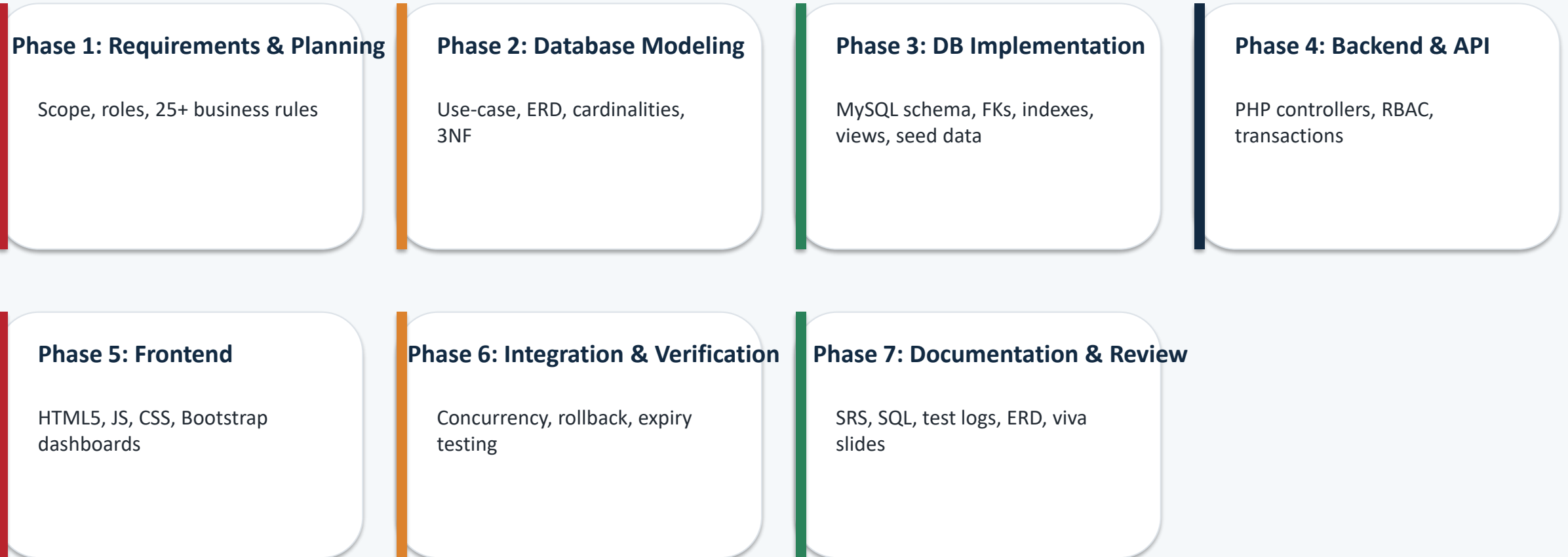
IN SCOPE

- End-to-end role-based coordination
- Individual blood-bag tracking with FEFO
- Rule-based donor sourcing
- ACID concurrency and transactions
- Inventory reallocation + audit history

OUT OF SCOPE

- Clinical or diagnostic testing
- Public exposure of donor phone/address
- AI/ML models
- Blockchain ledger
- Payment gateways
- Live GPS tracking

9. Methodology — Agile Iterative Lifecycle



10. Technology Stack

Frontend

HTML5 • CSS3 • JavaScript
Responsive UI

Backend

PHP 8.x
Business logic, session RBAC, MySQL communication

Database

MySQL
3NF relational storage, constraints, views, indexes, ACID transactions

Local Server

XAMPP (Apache + MySQL)
Cross-platform local development environment

Tools & VCS

VS Code • Git • GitHub
Coding, version control, SQL/schema management, collaboration

11. Expected Deliverables

- Centralized Relational Schema:** Fully normalized MySQL database (**3NF**) with 18 relational tables, PKs, FKs, CHECK constraints, and indexes.
- Individual Blood Bag Tracking:** Unique Bag IDs with collection dates, expiry dates, storage locations, and operational statuses.
- Role-Based Web Portals:** Dedicated dashboards for **Donors, Approved Hospital Staff, and Administrators**.
- Rule-Based Sourcing Engine:** Evaluate blood compatibility, inventory availability, donor eligibility, and proximity/priority.
- Transactional Fulfillment Workflow:** Support atomic reservations, issuing, rollback on concurrency failure, and inventory movement logs.
- Administrative Analytics & Auditing:** Provide inventory alerts, demand forecasts, near-expiry alerts, and complete system audit logs.

12. SQL Coverage & Application Mapping

DML

Requirement: SELECT, INSERT, UPDATE, DELETE

Implementation: CRUD across registration, profiles, bags and dashboards

Aggregation

Requirement: COUNT, SUM, AVG, MIN, MAX + GROUP BY/HAVING

Implementation: Blood totals and hospital monthly consumption

Joins

Requirement: INNER JOIN + LEFT JOIN

Implementation: Hospital dashboards; donor statistics including zero history

Subqueries

Requirement: Correlated / nested subquery

Implementation: Eligible donor sourcing based on group + 90-day interval

Views

Requirement: vw_available_inventory

Implementation: Non-expired, unreserved bags for FEFO queries

Transactions

Requirement: BEGIN / COMMIT / ROLLBACK

Implementation: Lock bags → fulfillment → reserve → request status → commit/rollback

The diagram illustrates the database structure for a blood donation management system. It consists of the following tables and their attributes:

- ROLES**: role_id (PK), role_name, description
- USERS**: user_id (PK), role_id (FK), username, email, phone, account_status
- SYSTEM_SETTINGS**: setting_id (PK), updated_by (FK), setting_key, setting_value
- AUDIT_LOGS**: audit_id (PK), user_id (FK), action, timestamp
- RECOGNITION_LEVELS**: level_id (PK), level_name, minimum_donations
- HOSPITAL_STAFF**: staff_id (PK), user_id (FK), hospital_id (FK), designation, staff_status
- HOSPITALS**: hospital_id (PK), approved_by (FK), hospital_name, registration_number, city, approval_status
- DONORS**: donor_id (PK), user_id (FK), blood_group_id (FK), full_name, city, availability_status, eligibility_status
- BLOOD_GROUPS**: blood_group_id (PK), blood_group_name
- DONATIONS**: donation_id (PK), donor_id (FK), recorded_by (FK), donation_date, screening_status, donation_status
- BLOOD_REQUESTS**: request_id (PK), hospital_id (FK), requested_by (FK), requested_type, urgency, status
- BLOOD_BAGS**: blood_bag_id (PK), donation_id (FK), storage_location_id (FK), bag_number, status, expiry_date
- FULFILLMENTS**: fulfillment_id (PK), request_id (FK), fulfilled_by (FK), fulfillment_status, total_quantity
- REQUEST_ITEMS**: request_item_id (PK), request_id (FK), blood_group_id (FK), quantity_requested, quantity_reserved
- INVENTORY_MOVEMENTS**: movement_id (PK), blood_bag_id (FK), from_location_id (FK), to_location_id (FK), performed_by (FK), movement_type, quantity
- FULFILLMENT_ITEMS**: fulfillment_item_id (PK), fulfillment_id (FK), blood_bag_id (FK), request_item_id (FK), quantity
- DONOR_MATCHES**: match_id (PK), request_id (FK), request_item_id (FK), donor_id (FK), match_status
- DONOR_RECOGNITION**: recognition_id (PK), donor_id (FK), level_id (FK), awarded_by (FK), achievement_date
- NOTIFICATIONS**: notification_id (PK), user_id (FK), request_id (FK), title, is_read

Key relationships include:

- ROLES** to **USERS**: one-to-many relationship.
- USERS** to **HOSPITALS**: one-to-many relationship (approved_by).
- USERS** to **DONORS**: one-to-many relationship.
- USERS** to **DONATIONS**: one-to-many relationship (recorded_by).
- USERS** to **BLOOD_REQUESTS**: one-to-many relationship (requested_by).
- USERS** to **INVENTORY_MOVEMENTS**: one-to-many relationship (performed_by).
- USERS** to **NOTIFICATIONS**: one-to-many relationship.
- HOSPITALS** to **DONATIONS**: one-to-many relationship.
- HOSPITALS** to **BLOOD_REQUESTS**: one-to-many relationship.
- HOSPITALS** to **BLOOD_BAGS**: one-to-many relationship (storage_location_id).
- DONORS** to **DONATIONS**: one-to-many relationship.
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- DONATIONS** to **BLOOD_BAGS**: one-to-many relationship.
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14. Schema — Relational Design

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15. Key DBMS Workflow — Emergency Fulfillment Transaction



Concurrency Protection

Row-level locking + ACID transaction prevents two hospitals from allocating the same blood unit simultaneously.

Failure Handling

If a concurrency or validation failure occurs, the workflow performs ROLLBACK so partial changes are not persisted.

16. Risks & Dependencies

Concurrency & Race Conditions

Risk: simultaneous requests reserve identical units.

Mitigation: strict DB transactions + row-level locking.

Data Privacy & Security

Risk: unauthorized exposure of donor PII.

Mitigation: RBAC and system-mediated notifications.

User Adoption / Response

Dependency: emergency matching depends on prompt donor responses.

Mitigation: maintain clear FEFO buffer stocks.

Medical Eligibility Variations

Dependency: external regulations may change eligibility intervals.

Mitigation: model intervals as configurable database parameters.

Academic / Technical Dependency

Successful delivery depends on correct schema implementation, testing, integration and documentation across all 7 Agile phases.

17. Conclusion

BloodLink transforms fragmented blood management into a centralized, reliable relational database system.

Strong Q emphasis: 3NF schema, constraints, indexes, views, joins, aggregation, subqueries and ACID transactions.

FEFO and rule-based donor matching directly address expiry, shortage and emergency sourcing challenges.

Role-based portals and audit trails support controlled, transparent coordination among donors, hospitals and administrators.

Future implementation will proceed through the defined Agile phases, testing and final documentation.

Thank You — Questions & Discussion